ML & AI FOR LIFE SCIENCES 2026 · DAY 1 · LECTURE 1

What AI/ML can and cannot do

…and how to look at your data first: PCA & UMAP



Trần Thượng Vân Du · SIB Swiss Institute of Bioinformatics

VNUHCM-University of Science · 02/11/2026

ROADMAP

VI: lộ trình

Three days, one question: what can the data tell us?

DAY 1 · FOUNDATIONS

- **Capabilities, limits; PCA & UMAP** ← now
- Unsupervised learning
- Linear regression
- Logistic regression & trees

DAY 2 · MORE POWER

- Random forest & XGBoost
- Neural networks, autoencoders, CNN
- LLMs as research assistants

DAY 3 · RESPONSIBILITY

- AI ethics: fairness, privacy, bias
- Capstone in mixed teams
- Presentations & feedback

WHY NOW?

VI: dữ liệu lớn

Biology now produces more data than people can read

~$100M → <$1,000

Cost of sequencing one human genome, 2001 → today

200M+

Predicted protein structures in the AlphaFold database

~500,000

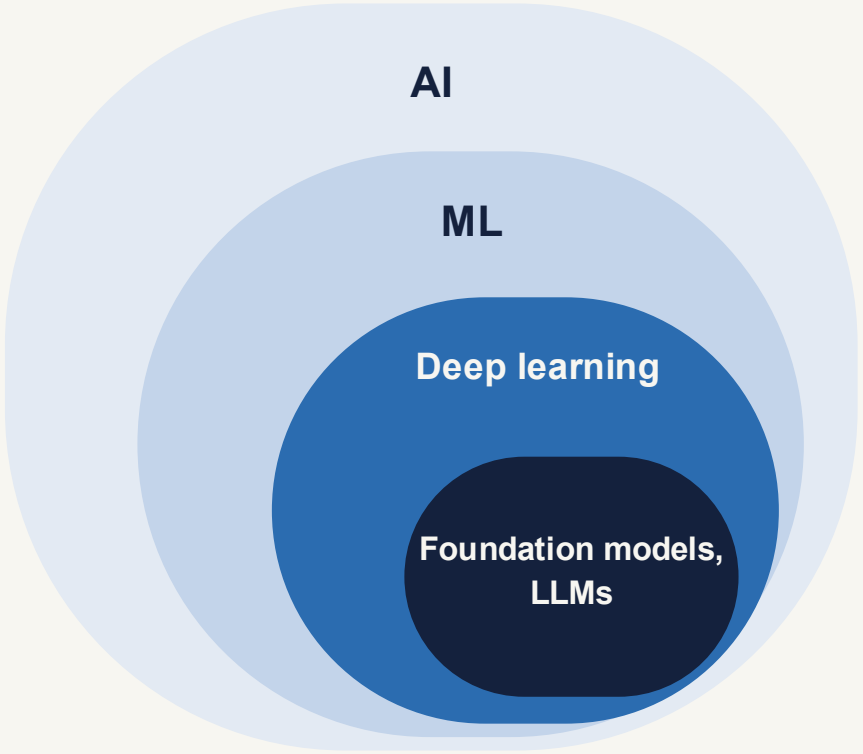
UK Biobank participants with whole-genome sequences

Sources: NHGRI sequencing cost data; AlphaFold DB (EMBL-EBI); UK Biobank WGS release 2023

VOCABULARY

VI: trí tuệ nhân tạo · học máy · học sâu

AI, ML, deep learning: nested ideas



AI — any system that performs tasks we associate with intelligence (rules included)

ML — systems that *learn the rules from data* instead of being given them

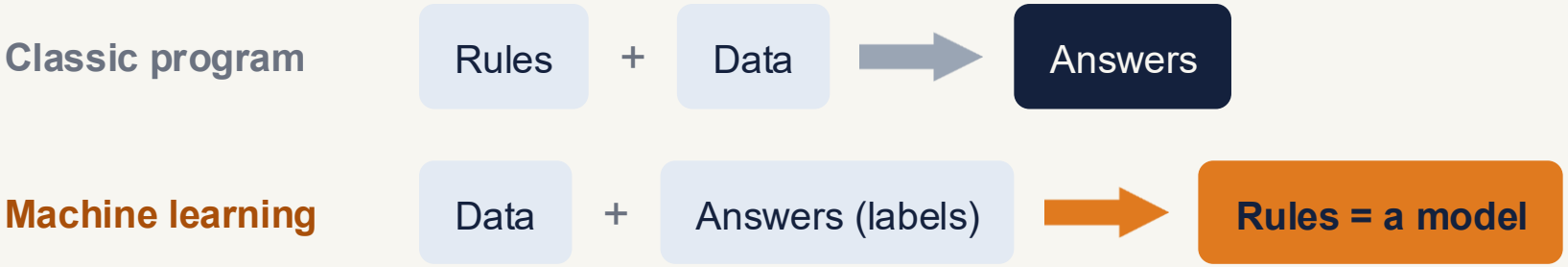
Deep learning — ML with many-layered neural networks (Day 2)

Foundation models — very large models pre-trained on huge data, then adapted (AlphaFold, ESM, ChatGPT, Claude)

THE CORE IDEA

VI: mô hình học từ dữ liệu

Programming writes the rules; ML learns them



Then we apply the learned model to **new** samples. The only question that matters: **does it work on data it has never seen?**

A DEFINITION YOU CAN USE

VI: nhiệm vụ · kinh nghiệm · hiệu năng

Every ML problem has a Task, Experience and Performance

T

Task

What should the model do?
Find splice sites in a genome sequence

E

Experience

What data does it learn from?
Sequences with known, annotated splice sites

P

Performance

How do we measure success?
% of true splice sites found in held-out chromosomes

SHARED LANGUAGE · BIOLOGY ⇄ CS

VI: ma trận dữ liệu · mẫu · đặc trưng · nhãn

Almost everything this week is one matrix X and a label y

sample	TP53	ESR1	…	MYC	y
patient_01	812	3	…	240	ER-
patient_02	655	4120	…	98	ER+
patient_03	903	2875	…	131	ER+
…	…	…	…	…	…

n rows (samples) × p columns (features) + one label column

Row = sample: a patient, a cell, a tissue biopsy

Column = feature: a gene, a CpG site, a metabolite

Value often = read counts: skewed, so we **log-transform** and normalise for sequencing depth

y = label: what we want to predict (may be absent)

MAP OF THE FIELD

VI: có giám sát · không giám sát · tăng cường

Four ways a machine can learn

Type	Question it answers	Biology example	In this workshop
Unsupervised	What structure is hidden in X?	Cell types in scRNA-seq	L1–L2 today
Supervised	Given X, predict a known y	Tumour subtype from expression	L3–L5
Self-supervised	Predict a hidden part of the data itself	Protein language models (ESM)	Day 2: DL, LLMs
Reinforcement	Which action maximises reward?	Molecule design loops	Mention only

Rule of thumb: **no label → unsupervised; label → supervised**. Number label → regression; class label → classification.

PART 2

What ML does well

Four results that changed how biology is done

CAPABILITIES

VI: năng lực của AI

ML already solves real problems in life science

Protein structure

AlphaFold 2 predicts 3D structure from sequence at near-experimental accuracy; Nobel Prize in Chemistry 2024.

Variant interpretation

AlphaMissense scores all ~71 million possible human missense variants as likely benign or pathogenic.

Digital pathology

Deep learning predicts microsatellite instability in colorectal cancer directly from H&E slides.

Single-cell atlases

Automatic cell-type annotation (e.g. CellTypist, scVI) across millions of cells and many studies.

WHEN TO USE ML

VI: khi nào nên dùng học máy?

ML shines when four conditions hold

Good fit

- Many examples, reliably labelled
- The pattern is stable over time and sites
- Prediction is the goal, not mechanism
- Errors are cheap to check downstream

Poor fit

- 30 samples, 20,000 features
- Training data from one hospital or one batch
- You need a causal answer ("does X cause Y?")
- A wrong answer harms a patient directly

PART 3

…and where it fails

A model learns whatever pattern predicts best, including the wrong ones

LIMIT 1 · CORRELATION IS NOT CAUSATION

VI: tương quan ≠ nhân quả

A model learned that asthma lowers pneumonia death risk



In the training data, asthmatic pneumonia patients **were treated more aggressively**, so they died less. The pattern was real; the conclusion "send asthmatics home" would be deadly.

Prediction ≠ explanation. Coefficients and feature importances describe the data, not the biology.

Caruana et al. KDD 2015 (pneumonia risk models)

COVID-19 X-ray models learned the hospital, not the lungs

Positive and negative images came from **different sources**. Models used text markers, patient position and image borders: features that told them the *source*, not the disease.

Accuracy was high inside the dataset and **dropped sharply on images from new hospitals**.

The same trap in omics

- Cases sequenced in 2023, controls in 2024
- Tumours from hospital A, normals from B
- One lab's protocol for one subtype

The model will learn the batch.

DeGrave, Janizek & Lee, Nat Mach Intell 2021 · Zech et al. PLoS Med 2018

LIMIT 3 · BIOLOGICAL DATA IS DIFFERENT

VI: đặc thù dữ liệu sinh học

Eight reasons textbook ML breaks on biology

$p \gg n$
20,000 genes, 80 patients

Batch effects
Date, lab, kit, operator

Confounders
Age, sex, BMI, medication

Replicates
Technical ≠ biological

Grouped samples
Many cells or visits per patient

Imbalance
Rare disease: 1 case in 100

Noisy labels
Pathologists disagree

Odd distributions
Sparse counts, compositional microbiome

These matter more than the choice of algorithm.

GARBAGE IN, GARBAGE OUT

A model is never better than its data and its question.

So before any model: **look at the data.**

PART 4

Exploring data with PCA & UMAP

Seeing 20,000 dimensions on a 2D screen

Six checks before any model

1 · Size

How many samples per group? Per batch?

2 · Missing values

Where, how many, and missing why?

3 · Distributions

Skewed? Needs log? Zeros?

4 · Scale

Features in very different units?

5 · Outliers

Biology or a failed sample?

6 · Metadata

Colour every plot by batch, site, sex, date

With 20,000 genes, every sample is far from every other

20,000

features (genes)

100

samples (patients)

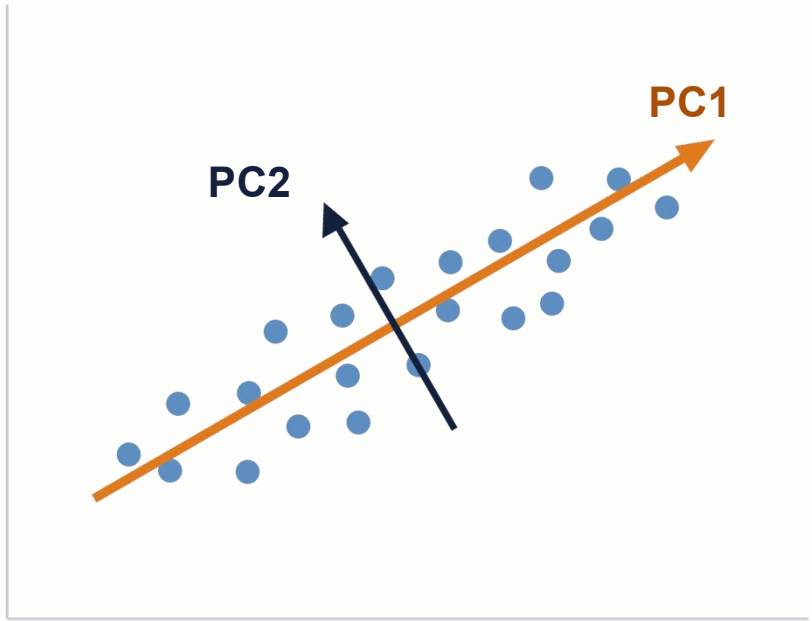
We cannot plot 20,000 axes.

In high dimensions, distances become similar and noise dominates.

But genes are **correlated**: pathways move together. The real information lives in far fewer dimensions.

Dimensionality reduction finds those few dimensions.

PCA rotates the axes to follow the spread of the data



PC1 = the direction of largest variance: a weighted mix of all genes.

PC2 = the next largest, at right angles to PC1. And so on.

Keep the first few PCs → most of the signal in 2–50 dimensions instead of 20,000.

Linear, deterministic, fast. Always **centre** (and usually **scale**) features first.

VI: điểm số · phương sai giải thích · hệ số tải

Three outputs, three questions

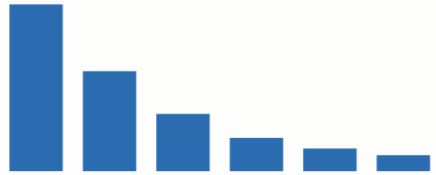
Scores

Where is each sample?

The PC1 vs PC2 scatter plot. Colour by every metadata column: groups, outliers, batches.

Explained variance

How much does each PC keep?



Scree plot: keep PCs before the "elbow".

Loadings

Which genes drive a PC?

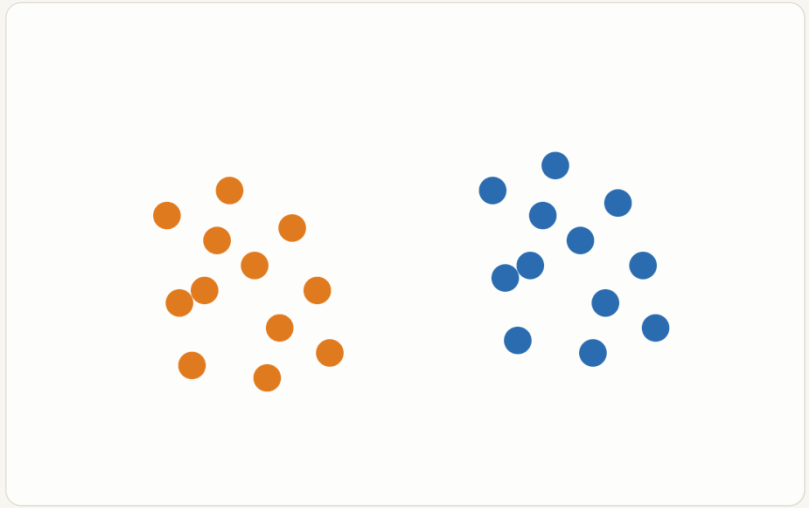
Top-weighted genes of PC1 → pathway enrichment → a biological name for the axis.

PCA AS A QUALITY-CONTROL TOOL

VI: hiệu ứng lô (batch effect)

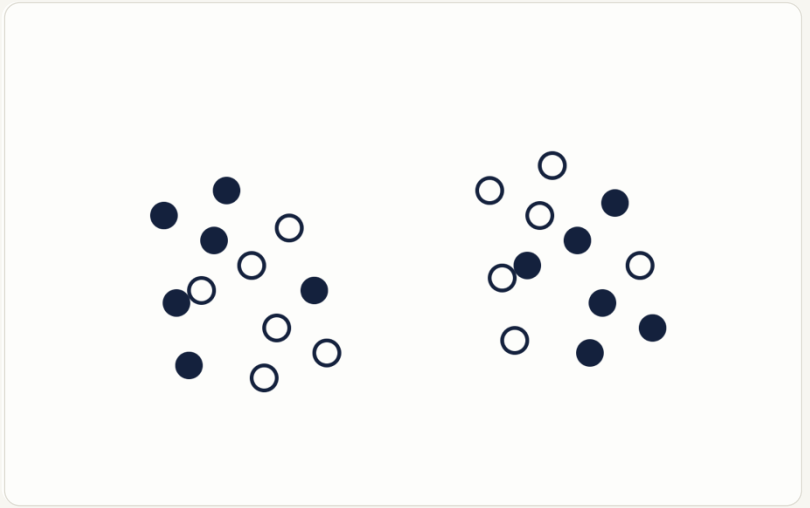
Same PCA plot, two colourings: which one explains PC1?

Coloured by sequencing batch



Orange = batch 1 · blue = batch 2 · **clear split**

Coloured by disease status



Filled = disease · hollow = control · **no split**

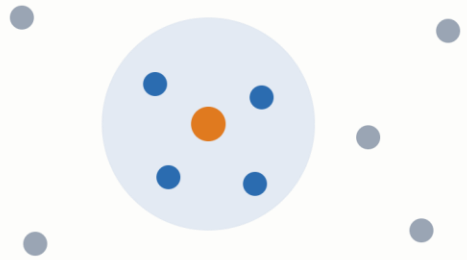
PC1 = batch, not biology. Fix the design or correct the batch **before** any modelling.

UMAP · THE INTUITION

VI: láng giềng gần nhất · đồ thị

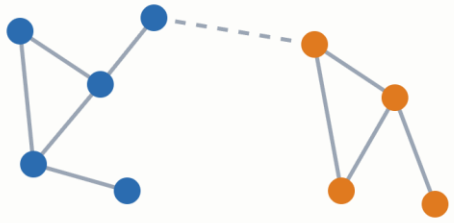
UMAP keeps neighbours together, and only neighbours

1 · Find neighbours



For each cell, its k nearest cells in the high-dimensional space (often in PCA space).

2 · Build a graph



Connect neighbours; strong edges for close cells, weak edges for far ones.

3 · Lay it out in 2D



Move points in 2D until the 2D graph looks like the original graph.

UMAP · READ WITH CARE

VI: diễn giải cẩn thận

A UMAP is a map of neighbourhoods, not of distances

You can read

- Which samples are local neighbours
- That groups probably exist (then verify)
- Where a marker gene or a batch sits

You cannot read

- Distances between clusters
- Cluster size or density
- Meaning of the axes

Knobs that change it

- n_neighbors: local ↔ global
- min_dist: how tightly packed
- Random seed

Rule: **explore with UMAP, conclude with statistics.**

SUMMARY

VI: so sánh PCA và UMAP

PCA and UMAP answer different questions

	PCA	UMAP
Type	Linear	Non-linear
Same result every run?	Yes	No (seed, parameters)
Axes mean something?	Yes: % variance, loadings	No
Preserves	Global structure	Local neighbourhoods
Best for	QC, batch, input to models	Visualising groups, e.g. cell types

In practice you use both: **PCA first**, then UMAP on the top PCs.

TAKE-AWAYS → COLAB

Three things to remember

1

ML learns patterns, including batch effects and shortcuts.

2

The real test is always new, independent data.

3

Look first: PCA for structure and QC, UMAP for neighbourhoods.